DandoEv — Third-Party Tools & Production Pricing Architecture

Comprehensive infrastructure breakdown, unit costs, and operational budget by active user volume.

Core Stack Architecture: Unified Open Banking (Mono) for KYC, 6-month statement analysis, CBN credit bureau default records, and direct debit mandates, with multi-channel messaging (Termii & Resend) and secure cloud infrastructure.

1. Third-Party Integration Matrix & Pricing

Service & Category	Provider	Operational Purpose	Pricing Model & Rates
Open Banking & Credit Suite Core Fintech Engine	**Mono Technologies**	• Bank linking & BVN/NIN lookup • 6-Mo statement cashflow scoring • Active loan detection across banks • CBN Credit Bureau default history • DirectPay & Recurring Mandates	• BVN/KYC: ₦50 – ₦100 / check • Statement Analysis: ₦100 – ₦150 / acct • Credit Bureau Check: ₦200 – ₦350 / check • DirectPay: 1% (capped at ₦1,000) • Mandate Debit: ~₦50 / debit
Card & Virtual Accounts Payment Gateway (Fallback)	**Paystack**	• Down payment card checkouts • Reusable card tokenization fallback • Automated bank settlements	• Local Transactions: 1.5% + ₦100 (₦100 waived for < ₦2,500), capped at ₦2,000 • Int'l Cards: 3.9% + ₦100 • Settlement: Free / ₦10 flat
SMS & WhatsApp Communications	**Termii**	• 2FA OTP verification SMS • Daily savings & repayment alerts • WhatsApp customer notices	• Local SMS (DND Route): ₦4.00 – ₦5.50 / SMS • WhatsApp Message: ₦6.50 – ₦8.50 / msg • Sender ID: Free / ₦5,000 regulatory setup
Transactional Email Communications	**Resend**	• Loan contract agreements (PDF) • Payment receipts • Security & password alerts	• Free Tier: 3,000 emails/mo • Pro Plan: $20/mo (~₦30,000) for 50,000 emails ($1.20 / add'l 1k)
Encrypted Storage Cloud Media	**Cloudinary / AWS S3**	• Encrypted KYC ID document uploads • E-Bike product images	• AWS S3: $0.023/GB (~$5 – $15/mo) • Cloudinary: Free tier (25 credits) or $99/mo Plus plan
Cache & BullMQ Queues Job Queue / Cache	**Upstash / Redis**	• Automated daily savings sweeps • API rate limiting • Reconciliation workers	• Free Tier: 10,000 commands/day • Pay-As-You-Go: $0.20 per 100k commands (~$10 – $25/mo)
Primary Database Data Store	**Railway / Supabase**	• PostgreSQL 16 relational DB • Automated backups & pooling	• Starter / Dev: $5 – $10/mo • Production Pro: $25 – $50/mo (~₦38,000 – ₦75,000/mo)
App & Edge Hosting Deployment	**Vercel**	• Next.js 16 App Router hosting • Edge CDN & SSL certificates	• Pro Plan: $20 / team seat / month (~$20 – $40/mo)
Error & APM Tracing Observability	**Sentry.io**	• Crash monitoring & API latency • Performance tracing	• Developer Tier: Free (5,000 errors/mo) • Team Plan: $26/mo (~₦39,000/mo)

2. Estimated Monthly Operational Budget by Active User Volume

Active User Range	Active Financed Bikes	Variable API Fees (Mono + Termii)	Fixed Infrastructure (DB + Vercel + Email)	Total Est. Monthly
50 – 100 Active Users	50 – 100 bikes	~₦45,000 – ₦90,000 (Mono KYC & Mandates + SMS)	~$70 / mo (~₦105,000) (Vercel, DB, Redis, Resend)	**₦150,000 – ₦195,000 (~$100 – $130 / mo)**
100 – 500 Active Users	100 – 500 bikes	~₦90,000 – ₦350,000 (Daily sweeps + SMS alerts)	~$100 / mo (~₦150,000) (Production DB & Redis)	**₦240,000 – ₦500,000 (~$160 – $330 / mo)**

500 – 1,000 Active Users	500 – 1,000 bikes	~₦350,000 – ₦700,000 (High-frequency mandate debits)	~$160 / mo (~₦240,000) (High-capacity Pro DB)	**₦590,000 – ₦940,000 (~$390 – $625 / mo)**
1,000 – 5,000 Active Users	1,000 – 5,000 bikes	~₦700,000 – ₦2,800,000 (Volume tiered mandate discounts)	~$300 / mo (~₦450,000) (Multi-instance Redis & DB)	**₦1,150,000 – ₦3,250,000 (~$765 – $2,160 / mo)**
5,000+ Active Users	5,000+ bikes	~₦2,800,000+ (Custom enterprise volume rates)	~$500 / mo (~₦750,000) (Enterprise HA cluster)	**₦3,550,000+ (~$2,360+ / mo)**

 Cost Optimization & Margin Strategy:
1. Amortize Processing Fees: DandoEv's standard ₦20,000 one-time origination fee on loan creation covers all customer onboarding API expenses (Mono KYC + Statement analysis + Credit check = ~₦500 total cost per applicant), delivering immediate net revenue.
2. WhatsApp vs. SMS: Route non-urgent repayment reminders via WhatsApp (Termii) for ~₦6.50/msg with 98% delivery rate without DND SMS blocking.
3. Capped Payment Gateway Fees: Both Mono DirectPay (capped at ₦1,000) and Paystack (capped at ₦2,000) protect margins on large down payment checkouts of ₦150,000 – ₦400,000+.